# Business Intelligence Internship – Data Simulation & Analytics Assignment

**Dataset Description**

Table 1: Policy Sales Data

| Column | Description |
|---|---|
| Customer_ID | Unique customer |
| Vehicle_ID | Unique vehicle |
| Vehicle_Value | ₹100000 |
| Premium | ₹100 × policy tenure |
| Policy_Purchase_Date | Vehicle purchase date in 2024 |
| Policy_Start_Date | Purchase date + 365 days |
| Policy_End_Date | Start date + tenure |
| Policy_Tenure | 1, 2, 3, or 4 years |

This is the description of policy_sales data. And I used PosgreSQL.

The table query given below:

```
CREATE TABLE policy_sales_data (
    customer_id BIGINT PRIMARY KEY,
    vehicle_id BIGINT,
    vehicle_value NUMERIC(10,2),
    premium NUMERIC(10,2),
    policy_purchase_date DATE,
   policy_start_date DATE,
    policy_end_date DATE,
    policy_tenure INT
);
SELECT * FROM policy_sales_data;
```

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Dataset Description

Table 2 claims data

| Column | Description |
|---|---|
| Claim_ID | Unique claim |
| Customer_ID | Customer |
| Vehicle_ID | Vehicle |
| Claim_Amount | ₹10,000 (10% of vehicle value) |
| Claim_Date | Claim date |
| Claim_Type | 1 = First Claim, 2 = Second Claim |

```
CREATE TABLE claims_data (
   claim_id BIGINT PRIMARY KEY,
   customer_id BIGINT,
   vehicle_id BIGINT,
   claim_amount NUMERIC(10,2),
   claim_date DATE,
   claim_type INT
);
SELECT * FROM claims_data;
```

1.-- Calculate the total premium collected during the year 2024.

For this question, I used SQL to calculate the total premium collected during 2024. I filtered the policies purchased in 2024 and applied the **SUM function on the premium column** to get the

total amount collected by the company in that year. This helps understand the overall revenue generated from policy sales in 2024.

This is the query :

```
SELECT SUM(premium) AS total_premium_2024

FROM policy_sales_data

WHERE policy_purchase_date BETWEEN '2024-01-01' AND '2024-12-31';
```

2. --Calculate the total claim cost for each year (2025 and 2026) with a monthly breakdown.

To solve this, I used SQL to group the claim data by **year and month of the claim date**. Then I calculated the **total claim amount using SUM** for each month. This allowed me to see how much claim cost occurred in each month of 2025 and 2026 and helped analyze the trend of claims over time.

This is the Query:

```
SELECT

   EXTRACT(YEAR FROM claim_date) AS claim_year,

   EXTRACT(MONTH FROM claim_date) AS claim_month,

   SUM(claim_amount) AS total_claim_cost

FROM claims_data

WHERE claim_date BETWEEN '2025-01-01' AND '2026-12-31'

GROUP BY claim_year, claim_month

ORDER BY claim_year, claim_month;
```

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3.-- Calculate the claim cost to premium ratio for each policy tenure (1, 2, 3, and 4 years).

In this question, I calculated the **claim-to-premium ratio for each policy tenure (1, 2, 3, and 4 years)** using SQL. I grouped the data by policy tenure, summed the total premium and total claims for each tenure, and then calculated the ratio. This ratio helps identify which tenure is more profitable for the company.

This is the query:

```
SELECT
    p.policy_tenure,
    SUM(c.claim_amount) AS total_claim_cost,
    SUM(p.premium) AS total_premium,
    SUM(c.claim_amount) / SUM(p.premium) AS claim_to_premium_ratio
FROM policy_sales_data p
LEFT JOIN claims_data c
ON p.vehicle_id = c.vehicle_id
GROUP BY p.policy_tenure
ORDER BY p.policy_tenure;
```

4.-- Calculate the claim cost to premium ratio by the month in which the policy was sold

--(January–December 2024).

For this analysis, I used SQL to group policies by the **month in which they were sold in 2024**. Then I compared the total claims associated with those policies against the total premium collected for that month. This helps understand if policies sold in certain months have higher claim costs relative to the premium.

This is the query:

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```
SELECT
   EXTRACT(MONTH FROM p.policy_purchase_date) AS sale_month,
   SUM(c.claim_amount) AS total_claim_cost,
   SUM(p.premium) AS total_premium,
   SUM(c.claim_amount) / SUM(p.premium) AS claim_premium_ratio
FROM policy_sales_data p
LEFT JOIN claims_data c
ON p.vehicle_id = c.vehicle_id
GROUP BY sale_month
ORDER BY sale_month;
```

5.-- If every vehicle that has not yet made a claim eventually files exactly one claim during the

--remaining policy tenure, estimate the total potential claim liability.

In this question, I identified vehicles that **have not yet filed a claim** using SQL. Since the assumption is that every remaining vehicle will eventually make exactly one claim, I multiplied the number of such vehicles by the **standard claim amount** to estimate the company's total potential future claim liability

This is the query:

```
SELECT
   COUNT(p.vehicle_id) - COUNT(DISTINCT c.vehicle_id) AS vehicles_without_claim,
   (COUNT(p.vehicle_id) - COUNT(DISTINCT c.vehicle_id)) * 10000
   AS potential_claim_liability
```

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```
FROM policy_sales_data p
LEFT JOIN claims_data c
ON p.vehicle_id = c.vehicle_id;
```

6.-- Assume daily premium = Total Premium ÷ Total Policy Tenure Days. Based on this:

•-- Calculate the premium already earned by the company up to February 28, 2026.

•-- Estimate the premium expected to be earned monthly for the remaining policy period

--(assume 46 months remaining).

For this part, I first calculated the **daily premium by dividing total premium by the total policy tenure days** using SQL. Then I calculated how much premium has already been earned by the company **up to February 28, 2026** based on the number of days passed. Finally, I estimated the **monthly premium expected to be earned in the remaining 46 months**, which helps project the company's future revenue.

This is the query:

```
SELECT
SUM(
  premium / (policy_tenure * 365) *
  GREATEST(
    0,
    LEAST(policy_end_date, '2026-02-28') - policy_start_date
  )
```

```
) AS earned_premium
FROM policy_sales_data
WHERE policy_start_date <= '2026-02-28';
```

# Bonus questions

```
--1. Identify which policy tenure appears most profitable and explain why
SELECT
  --1. Identify which policy tenure appears most profitable and explain why?
```

To identify the most profitable policy tenure, I compared the **total premium collected** with the **total claim cost** for each tenure (1-year, 2-year, 3-year, and 4-year). I calculated the **loss ratio**, which is claim cost divided by premium. The tenure with the **lowest loss ratio is considered the most profitable**, because the company pays fewer claims relative to the premium it collects. This helps the insurer understand which policy duration generates better margins and lower risk. I solved this question with the help of postgresql.

The Query is given below:

```
SELECT
  p.policy_tenure,
    SUM(p.premium) AS total_premium,
    COALESCE(SUM(c.claim_amount),0) AS total_claims,
    COALESCE(SUM(c.claim_amount),0) / SUM(p.premium) AS loss_ratio
FROM policy_sales_data p
LEFT JOIN claims_data c
ON p.vehicle_id = c.vehicle_id
```

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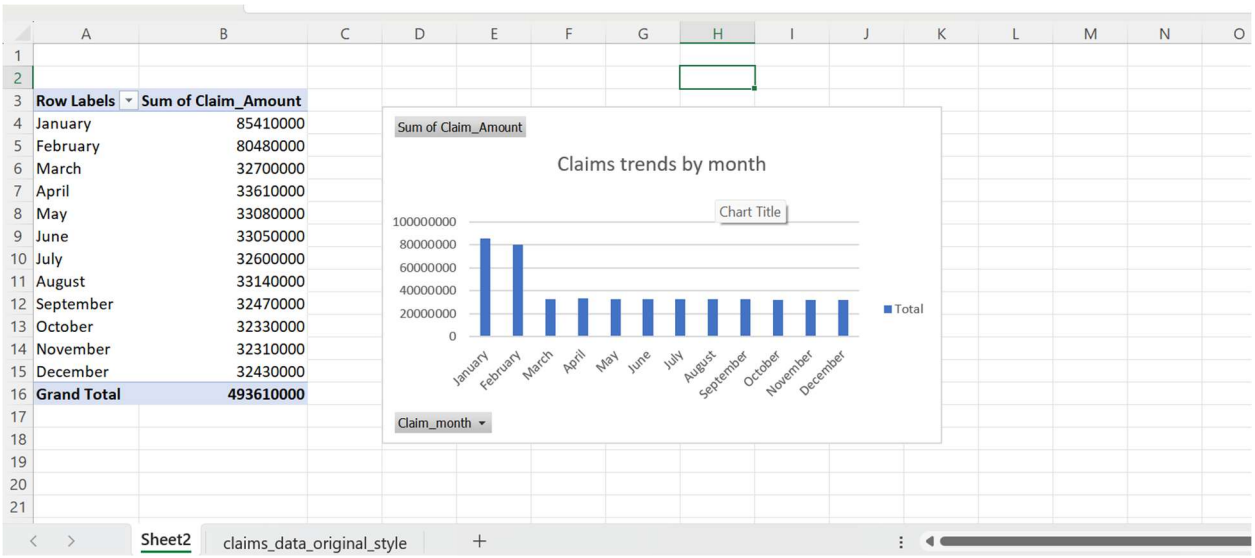
GROUP BY p.policy_tenure

ORDER BY loss_ratio;

--2. Build a simple dashboard or visualization showing claim trends by month.

To analyze claim trends, I created a **monthly visualization** using the claim date. I aggregated claims by month and displayed them using charts such as a **bar graph for total claim cost and a bar chart for number of claims**. This dashboard helps identify **seasonal patterns, spikes in claim activity, and periods with higher claim costs**, which can help the company monitor risk and plan reserves I used pivot table and pivot chart for this

Picture given below:



| Row Labels | Sum of Claim_Amount |
|---|---|
| January | 85410000 |
| February | 80480000 |
| March | 32700000 |
| April | 33610000 |
| May | 33080000 |
| June | 33050000 |
| July | 32600000 |
| August | 33140000 |
| September | 32470000 |
| October | 32330000 |
| November | 32310000 |
| December | 32430000 |
| **Grand Total** | **493610000** |



--3. Estimate the loss ratio (Claims ÷ Premium) for the portfolio.

The **loss ratio** measures the overall profitability of the insurance portfolio. It is calculated by dividing the **total claim amount by the total premium collected**. A lower loss ratio means the company is paying fewer claims relative to the revenue it earns from premiums, which indicates

better profitability. This metric is widely used in insurance to evaluate **financial performance and risk exposure**.

Query is given below:

```
SELECT
   SUM(c.claim_amount) AS total_claims,
   SUM(p.premium) AS total_premium,
   SUM(c.claim_amount) / SUM(p.premium) AS portfolio_loss_ratio
FROM policy_sales_data p
LEFT JOIN claims_data c
ON p.vehicle_id = c.vehicle_id;
```

--4. If claim frequency increases by 5% annually, estimate the impact on future profitability.

If claim frequency increases by 5% each year, the **total claim cost will also increase**, assuming the claim amount per claim remains the same. This would lead to a **higher loss ratio**, which reduces profitability for the insurance company. Over time, if premiums remain constant, the insurer may need to **adjust pricing, improve risk selection, or set aside higher reserves** to maintain profitability.

Query is given below:

```
SELECT
   SUM(claim_amount) AS current_claim_cost,
   SUM(claim_amount) * 1.05 AS next_year_claim_cost,
   SUM(claim_amount) * 1.10 AS_two_year_claim_cost
FROM claims_data;
```